

1. The fact that there are contingent things is contingent.
2. Every contingent fact has an explanation. (PSR)
3. The fact that there are contingent things has an explanation. (1,2)
4. The fact that there are contingent things can't be explained by any contingent thing.
5. The fact that there are contingent things is explained by some thing which is not contingent. (3,4)
6. The fact that there are contingent things is explained by some necessary thing. (5)
7. There is a necessary thing which explains the existence of contingent things. (6)
8. If there is a necessary thing which explains the existence of contingent things, then God exists.

C. God exists. (7,8)

THE COSMOLOGICAL ARGUMENT

We found reason to doubt the assumption that, if there is a first cause, then God exists. Do similar doubts apply to premise (8) of Leibniz's argument?

Here we can say things similar to the things we said about quasi-theism when we were discussing the first cause argument.

We can again consider the hypotheses of simple theism and simple atheism, and now consider alongside them the quasi-theistic hypothesis that the universe was created by a being with the unusual property that it is literally impossible for that being not to exist.

If this strikes you as a strange and unlikely hypothesis, then premise 8 of Leibniz's argument should strike you as a plausible one.



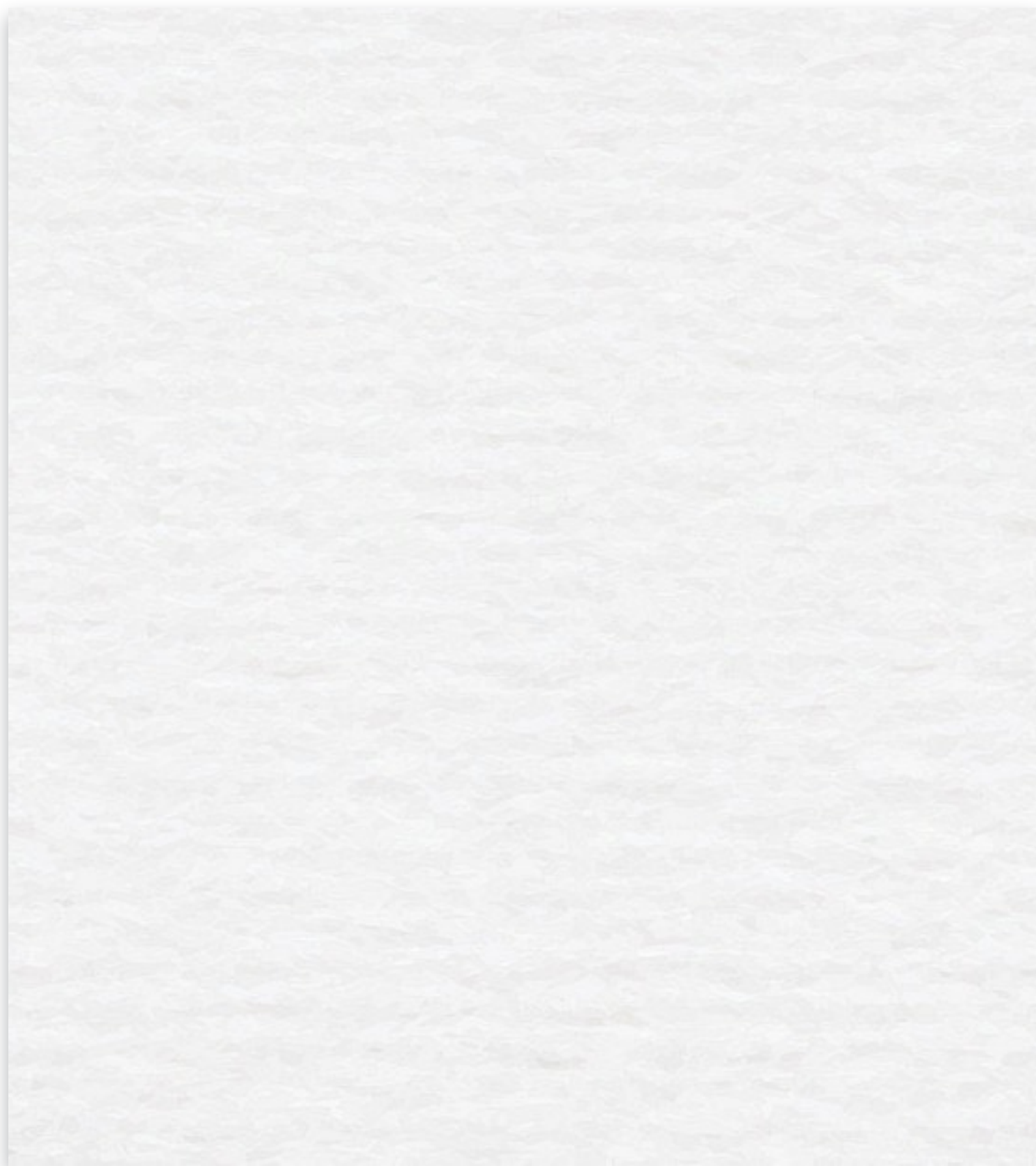
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... we cannot find in any of the individual things, or even in the entire collection and series of things, a sufficient reason for why they exist.

Let us suppose that a book on the elements of geometry has always existed, one copy always made from another. It is obvious that although we can explain a present copy of the book from the previous book from which it was copied, this will never lead us to a complete explanation, no matter how many books back we go, since we can always wonder why there have been such books.

What is true of these books is also true of the different states of the world ... however far back we might go into previous states, we will never find in those states a complete explanation for why there is a world at all, and why it is the way it is.



Leibniz's
argument



the
PSR



objections
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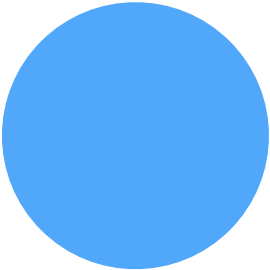
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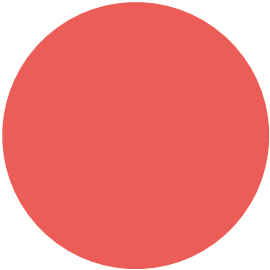
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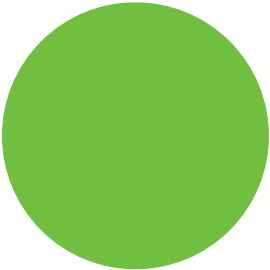
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the $\mathcal{H}_2$ norm of the error signal $\|e\|_2$ is

$$\|e\|_2^2 = \int_0^\infty e^T e \, dt = \int_0^\infty \begin{bmatrix} x & u \end{bmatrix} \begin{bmatrix} Q & 0 \\ 0 & R \end{bmatrix} \begin{bmatrix} x \\ u \end{bmatrix} dt \quad (10)$$

where Q and R are symmetric positive definite matrices.

The $\mathcal{H}_2$ norm of the error signal $\|e\|_2$ can be written as

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where Q and R are symmetric positive definite matrices.

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where Q and R are symmetric positive definite matrices.



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